

## Company History

Android began as Android, Inc., a company founded by Andy Rubin, Chris White, Nick Sears, and Rich Miner in October 2003. They focused on creating mobile devices that were able to take into account location information and user preferences. After successfully navigating market demand and financial difficulties, Google acquired Android, Inc., in August 2005. During the period following, Google began building partnerships with hardware, software, and telecommunications companies with the intent of entering the mobile market.

In November 2007, the Open Handset Alliance (OHA) was announced. This consortium of companies, which included 34 founding members led by Google, shares a commitment to openness. In addition, it aims to accelerate mobile platform innovation and offer consumers a richer, less expensive, and better mobile experience. The OHA has since grown to 84 members at the time this book was published. Members represent all parts of the mobile ecosystem, including mobile operators, handset manufacturers, semiconductor companies, software companies, and more. You can find the full list of members on the OHA website at [www.openhandsetalliance.com/oha\\_members.html](http://www.openhandsetalliance.com/oha_members.html).

With the OHA in place, Google announced its first mobile product, Android. However, Google still did not bring any devices running Android to the market. Finally, after a total of five years, Android was made available to the general public in October 2008. The release of the first publicly available Android phone, the HTC G1, marked the beginning of an era.

## Version History

Before the first commercial version of Android, the operating system had Alpha and Beta releases. The Alpha releases were available only to Google and OHA members, and they were codenamed after popular robots *Astro Boy*, *Bender*, and *R2-D2*. Android Beta was released on November 5, 2007, which is the date that is popularly considered the Android birthday.

The first commercial version, version 1.0, was released on September 23, 2008, and the next release, version 1.1, was available on February 9, 2009. Those were the only two releases that did not have a naming convention for their codename. Starting with Android 1.5, which was released on April 30, 2009, the major versions' code names were ordered alphabetically with the names of tasty treats. Version 1.5 was code named *Cupcake*. Figure 1-1 shows all commercial Android versions, with their respective release dates and code names.

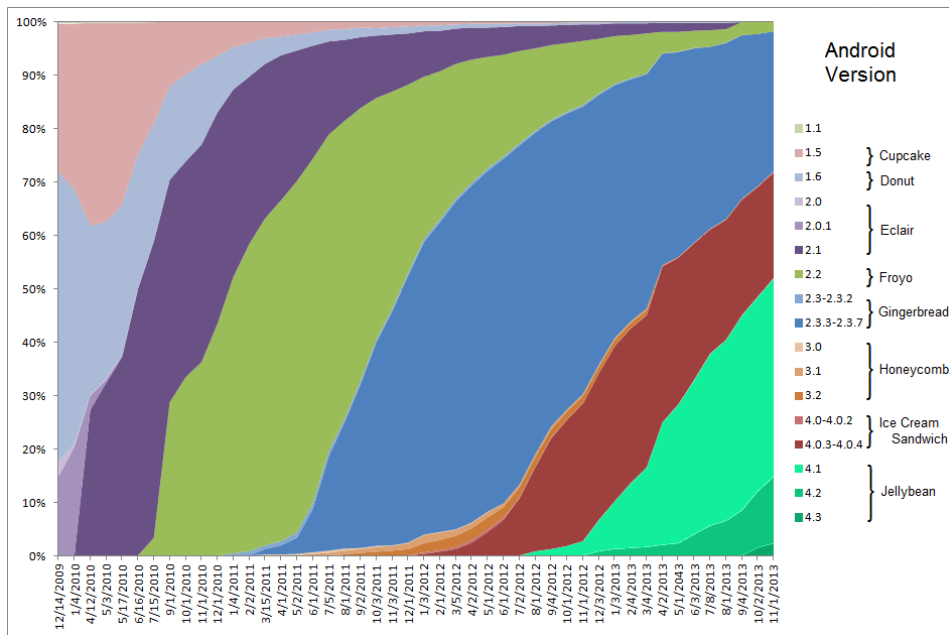
|      |         |      |  |  |  |  |  |  |  |  |
|------|---------|------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|      |         |      | Cupcake                                                                           | Donut                                                                             | Eclair                                                                            | Froyo                                                                             | Gingerbread                                                                       | Honeycomb                                                                          | Icecream sandwich                                                                   | Jelly Bean                                                                          |
| 2008 | 23, Sep | v1.0 |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
| 2009 | 9, Feb  | v1.1 |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 30, Apr |      | v1.5                                                                              |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 15, Sep |      |                                                                                   | v1.6                                                                              |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 26, Oct |      |                                                                                   |                                                                                   | v2.0                                                                              |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 3, Dec  |      |                                                                                   |                                                                                   | v2.0.1                                                                            |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
| 2010 | 12, Jan |      |                                                                                   |                                                                                   | v2.1                                                                              |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 20, May |      |                                                                                   |                                                                                   |                                                                                   | v2.2                                                                              |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 6, Dec  |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v2.3                                                                              |                                                                                    |                                                                                     |                                                                                     |
| 2011 | 18, Jan |      |                                                                                   |                                                                                   |                                                                                   | v2.2.1                                                                            |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 22, Jan |      |                                                                                   |                                                                                   |                                                                                   | v2.2.2                                                                            |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 9, Feb  |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v2.3.3                                                                            |                                                                                    |                                                                                     |                                                                                     |
|      | 22, Feb |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.0                                                                               |                                                                                     |                                                                                     |
|      | 28, Apr |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v2.3.4                                                                            |                                                                                    |                                                                                     |                                                                                     |
|      | 10, May |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.1                                                                               |                                                                                     |                                                                                     |
|      | 15, Jul |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.2                                                                               |                                                                                     |                                                                                     |
|      | 25, Jul |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v2.3.5                                                                            |                                                                                    |                                                                                     |                                                                                     |
|      | 2, Sep  |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v2.3.6                                                                            |                                                                                    |                                                                                     |                                                                                     |
|      | 20, Sep |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.2.1                                                                             |                                                                                     |                                                                                     |
|      | 21, Sep |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v2.3.7                                                                            |                                                                                    |                                                                                     |                                                                                     |
|      | 30, Sep |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.2.2                                                                             |                                                                                     |                                                                                     |
|      | 19, Oct |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | v4.0                                                                                |                                                                                     |
|      | 21, Oct |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | v4.0.1                                                                              |                                                                                     |
|      | 21, Nov |      |                                                                                   |                                                                                   |                                                                                   | v2.2.3                                                                            |                                                                                   |                                                                                    |                                                                                     |                                                                                     |
|      | 28, Nov |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | v4.0.2                                                                              |                                                                                     |
|      | 15, Dec |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.2.4                                                                             |                                                                                     |                                                                                     |
|      | 16, Dec |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | v4.0.3                                                                              |                                                                                     |
| 2012 | Jan     |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.2.5                                                                             |                                                                                     |                                                                                     |
|      | 15, Feb |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   | v3.2.6                                                                             |                                                                                     |                                                                                     |
|      | 29, Mar |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    | v4.0.4                                                                              |                                                                                     |
|      | 9, Jul  |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | v4.1                                                                                |
|      | 23, Jul |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | v4.1.1                                                                              |
|      | 9, Oct  |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | v4.1.2                                                                              |
|      | 13, Nov |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | v4.2                                                                                |
| 2013 | 27, Nov |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | v4.2.1                                                                              |
|      | 11, Feb |      |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     | v4.2.2                                                                              |

Figure 1-1: Android releases

of November 2013. Despite these small variations, all sources seem to agree that Android is the dominating mobile operating system.

### Release Adoption

Not all Android devices run the same Android version. Google regularly publishes a dashboard showing the relative percentage of devices running a given version of Android. This information is based on statistics gathered from visits to Google Play, which is present on all approved devices. The most up-to-date version of this dashboard is available at <http://developer.android.com/about/dashboards/>. Additionally, Wikipedia contains a chart showing dashboard data aggregated over time. Figure 1-3 depicts the chart as of this writing, which includes data from December 2009 to February 2013.



**Figure 1-3:** Android historical version distribution

Source: fjmustak (Creative Commons Attribution-Share Alike 3.0 Unported license) [http://en.wikipedia.org/wiki/File:Android\\_historical\\_version\\_distribution.png](http://en.wikipedia.org/wiki/File:Android_historical_version_distribution.png)

As shown, new versions of Android have a relatively slow adoption rate. It takes in excess of one year to get a new version running on 90 percent of devices. You can read more about this issue and other challenges facing Android in the “Grasping Ecosystem Complexities” section later in this chapter.

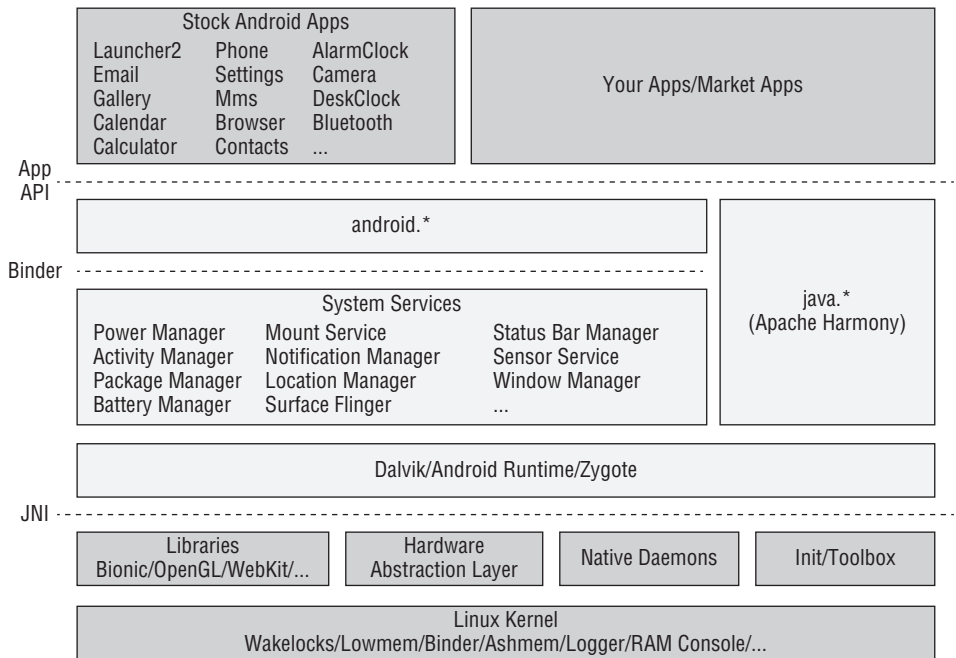
# Android Security Design and Architecture

Android is comprised of several mechanisms playing a role in security checking and enforcement. Like any modern operating system, many of these mechanisms interact with each other, exchanging information about subjects (apps/users), objects (other apps, files, devices), and operations to be performed (read, write, delete, and so on). Oftentimes, enforcement occurs without incident; but occasionally, things slip through the cracks, affording opportunity for abuse. This chapter discusses the security design and architecture of Android, setting the stage for analyzing the overall attack surface of the Android platform.

## Understanding Android System Architecture

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The general Android architecture has, at times, been described as “Java on Linux.” However, this is a bit of a misnomer and doesn’t entirely do justice to the complexity and architecture of the platform. The overall architecture consists of components that fall into five main layers, including Android applications, the Android Framework, the Dalvik virtual machine, user-space native code, and the Linux kernel. Figure 2-1 shows how these layers comprise the Android software stack.



**Figure 2-1:** General Android system architecture

Source: Karim Yaghmour of Opersys Inc. (Creative Commons Share-Alike 3.0 license)  
<http://www.slideshare.net/opersys/inside-androids-ui>

Android applications allow developers to extend and improve the functionality of a device without having to alter lower levels. In turn, the Android Framework provides developers with a rich API that has access to all of the various facilities an Android device has to offer—the “glue” between apps and the Dalvik virtual machine. This includes building blocks to enable developers to perform common tasks such as managing user interface (UI) elements, accessing shared data stores, and passing messages between application components.

Both Android applications and the Android Framework are developed in the Java programming language and execute within the Dalvik virtual machine (DalvikVM). This virtual machine (VM) was specially designed to provide an efficient abstraction layer to the underlying operating system. The DalvikVM is a register-based VM that interprets the Dalvik Executable (DEX) byte code format. In turn, the DalvikVM relies on functionality provided by a number of supporting native code libraries.

The user-space native code components of Android includes system services, such as vold and DBus; networking services, such as dhcpcd and wpa\_supplicant; and libraries, such as bionic libc, WebKit, and OpenSSL. Some of these services and libraries communicate with kernel-level services and drivers, whereas others simply facilitate lower-level native operations for managed code.